

Patterns for Prediction: An Audio-to-Audio Hidden Markov Framework

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Abstract—The Music Information Retrieval Evaluation eXchange (MIREX) Patterns for Prediction task evaluates the ability of computational models to generate accurate and stylistically coherent musical continuations. While prior approaches have predominantly relied on symbolic representations, this study investigates an end-to-end audio-to-audio framework to model pattern prediction using raw musical audio. Using the Patterns for Prediction Development Dataset (PPDD-Jul2018), the authors implemented a pipeline that processes tempo-normalized, synthesized audio into fixed temporal segments. The study evaluates four modeling strategies based on Hidden Markov Models (HMMs), comparing feature representations ranging from PCA-reduced spectrograms to high-level musical features in k-means cluster. Results demonstrate that a Variable-Order HMM trained on clustered musical features yields the most structurally coherent results. This configuration achieved a mean Cardinality Score of 0.263 and a Pitch Score of 0.429. Although the audio-based system currently underperforms symbolic Markov baselines, the research highlights the feasibility of extracting and extrapolating musical patterns directly from audio without symbolic intermediaries.

Index Terms—Music Information Retrieval (MIR), Patterns for Prediction, Audio Continuation, Hidden Markov Models (HMM), Audio-to-Audio Generation, Variable-Order Markov Models, Mel-spectrogram, Musical Feature Clustering.

I. INTRODUCTION

When listening to music, humans naturally form expectations about what will happen next [1]. The ability to anticipate forthcoming musical events contributes to the emotional engagement and reward associated with musical listening. In Music Information Retrieval (MIR), a range of algorithmic approaches has been developed to model this predictive capacity, particularly through systems designed to identify patterns and generate continuations. This paper focuses on the Music Information Retrieval Evaluation eXchange (MIREX) 2021 task Patterns for Prediction. The goal of this task is to evaluate how effectively computational models can learn from a given musical excerpt and produce a continuation that is both accurate and stylistically

coherent [2]. In alignment with this objective, the present study aims to simulate the human predictive process by training models that capture underlying musical patterns and anticipate plausible continuations. While prior work on the Patterns for Prediction task has relied mostly on symbolic musical representations [3], [4], this study investigates the possibility of creating an audio-to-audio pattern prediction systems. By exploring an end-to-end audio framework, we aim to model pattern prediction using raw musical audio rather than symbolic intermediaries.

II. LITERATURE REVIEW

Pattern discovery and prediction are central challenges in computational music analysis. Early methods focused on finding repeated motifs using string-based and geometric approaches. These relied on quantifiable metrics like compactness and compression ratio, but struggled in polyphonic or expressive music and often produced redundant results [5]. Recent improvements in motif detection have used more robust geometric methods, leading to better results in challenging contexts and helping cope with the variability of real musical data [6].

Traditional pattern prediction models use Markov Chain and Variable-Length Markov Chain (VLMC) models. These models rely on local context and often cannot capture longer-range patterns or adapt to complex musical structure. Multiple viewpoint systems, which combine several musical features for prediction, partly address these problems but still face scalability and data sparsity issues [7]. Distributed neural models, such as Restricted Boltzmann Machines [8], have improved performance by learning longer-term dependencies and scaling efficiently as sequence length increases, but even these have limits for real-world music.

Long-Short Term Memory (LSTM) neural networks brought a major advance. These models handle phrase continuation, long-term coherence, and style adaptation

far better than older methods [9], [10]. However, generating convincing large-scale musical structure remains a challenge for LSTMs, and they need large training datasets to reach their full potential.

The most recent shift is toward transformer-based architectures. Self-attention transformers, as seen in models like Music Transformer [11] and Jukebox [12], set a new standard for both pattern discovery and prediction. The Bar Transformer [13] introduced a hierarchical transformer framework that learns musical structure at both bar and track levels, using control tokens for features such as key, rhythmic density, and tonal tension to generate stylistically coherent and controllable compositions. These models capture long-range structure and can model motifs and stylistic details across entire pieces, outperforming LSTM and Markov models. They do require significant computational resources and large datasets, but surveys confirm that these deep learning methods now clearly outperform previous approaches in both discovery and generation tasks [14], [15].

However, nearly all existing approaches to the MIREX Patterns for Prediction task operate exclusively on symbolic musical representations. While symbolic models benefit from clear event structure and pattern boundaries, they bypass the acoustic detail, expressive nuance, and performance variability present in real audio. As a result, current systems do not address how predictive modeling might function in an audio-only setting. This gap is particularly notable given the recent success of deep learning architectures in audio generation and continuation tasks outside the MIREX framework. This study addresses that gap by working directly with audio, aiming for an end-to-end audio-to-audio system that learns and extrapolates musical patterns without relying on symbolic intermediaries. In doing so, it brings pattern-prediction research closer to real-world listening conditions.

III. METHODOLOGY

This project works directly with audio rather than symbolic MIDI to stay consistent with the audio-based MIREX Patterns for Prediction task. We investigated several Transformer-style and hybrid audio models, but they proved impractical due to missing implementations, unavailable pretrained weights, large data demands, and a stronger focus on creative generation than on interpretable, MIREX-style evaluation. We also considered alternatives such as parallel Markov models over mel bins and factorized pitch–timbre pipelines, but these either overlooked harmonic structure or collapsed into a symbolic continuation paradigm. These challenges motivated the exploration of audio-to-audio generation using a simple Hidden Markov Model (HMM). As no prior work addresses this approach, we present all of our experimentations on input representations and HMM

configurations leading to the best, proposed method (Figure 1).

To support reproducibility and further research, our main source code is available at <https://github.com/LingQi000809/6201PatternForPrediction>. The code for the first attempt is available at <https://github.com/Jamesprocode/Pattern-for-Prediction-audio-to-audio>.

A. Data

We use the PPDD-Jul2018 dataset derived from the Lakh MIDI Dataset [16]. Each entry contains a prime and continuation in both symbolic and audio form. To prepare the material for audio-domain pattern prediction, we use the small monophonic subset of the dataset and apply several preprocessing steps. Because the original primes exhibit varying tempos, all examples are first time-normalized to 120 BPM so that every observation attains the fixed temporal length and dimensionality required by the HMM framework. The tempo-normalized MIDI files are then synthesized in a sample rate of 22,050Hz using a fixed soundfont, providing a uniform timbre for feature extraction. Beat tracking is applied to validate temporal consistency. Finally, the audio excerpts are segmented into fixed 16th-note slices, providing inputs of uniform shape for model training. The 16th-note window is short enough to capture most individual notes, while longer notes span multiple consecutive slices. After filtering and verification, the final dataset contains 75 valid primes, of which 60 are used for training and 15 for testing.

B. Training Input Variants

We experiment with three different ways of representing the observations used for HMM training:

1) *PCA-reduced Raw Spectrograms*: We perform Principal Component Analysis (PCA) on the spectrogram slices ($n_{\text{mels}} = 80$) and reduce them to a compact 20-dimensional feature space. We aim to capture the major spectral variance with this lower-dimensional representation to enable efficient sequence modeling.

2) *Clustered Spectrograms*: We apply k-means clustering to the flattened spectrogram slices ($n_{\text{mels}} = 128$) to create discrete clusters. The optimal number of clusters is determined using the silhouette score, which quantifies how well each sample fits within its assigned cluster relative to neighboring clusters. Scores near +1 indicate compact, well-separated clusters, while scores close to 0 suggest overlapping structures. For the raw spectrograms, the highest mean silhouette score is achieved at $K = 290$, indicating the most stable clustering. The HMM is trained on sequences of cluster IDs and generates predicted cluster sequences, with each predicted observation mapped back to its cluster centroid for decoding into a spectrogram.

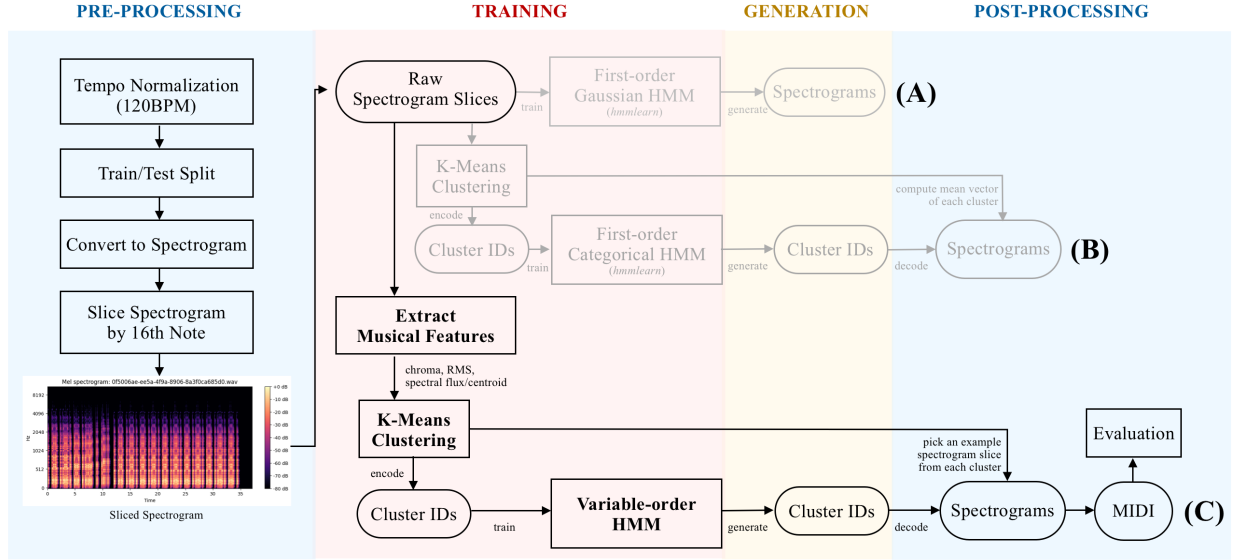


Fig. 1: Flowchart of the full pipeline and all modeling attempts:

(A) Attempt 1 - raw spectrograms with first-order Gaussian HMM: We trained a first-order HMM directly on raw spectrograms, which produced only noise-like outputs.

(B) Attempt 2 - clustered spectrograms with first-order Categorical HMM: We grouped raw spectrograms into 64 clusters and trained a first-order HMM on the resulting cluster-ID sequences. The generated cluster IDs were mapped back to spectrograms via cluster means. This yielded some melodic fragments but remained noisy and jittery.

(C) Attempt 3 (best) - clustered musical features with variable-order HMM: We extracted musical features before clustering, trained a variable-order HMM to capture longer context, and reconstructed spectrograms by sampling real examples from each cluster. This approach consistently produced melodic content.

3) Clustered Musical Features: We extract musical features from the spectrogram slices ($n_mels = 128$), including chroma, center pitch, and RMS amplitude. The goal is to group slices that share similar pitch and dynamic characteristics, so that each cluster represents a musically coherent type of observation. The resulting cluster IDs serve as discrete observations for HMM training. Based on silhouette scores computed over a range of K values, we select $K = 500$ for the feature sets. Instead of relying on cluster centroids—which can average out important spectral details—we experiment with a representative-instance strategy. We identify an exemplar spectrogram for each cluster that is nearest to the cluster center in Euclidean space. That exemplar spectrogram is used to reconstruct the predicted spectrogram.

C. HMM Model Variants

Three variants of Hidden Markov Models are used respectively for the three input representations:

1) Gaussian HMM for Continuous Features: For the first experiment, where inputs are PCA-reduced spectrograms, we employ a Gaussian Hidden Markov Model (HMM) with 10 hidden states from the `hmmlearn` library. In this model, each hidden state is associated with a multivariate Gaussian distribution of possible

observation vectors, so it is well suited for continuous spectrogram-based inputs.

2) Categorical HMM for Clustered Features: For the second experiment, observations are represented by k-means cluster IDs. We use a Categorical HMM from the `hmmlearn` library, in which each hidden state emits a discrete cluster ID according to a learned categorical distribution. This model is suitable when the feature space has been quantized into meaningful musical events, making continuous Gaussian emissions unnecessary.

3) Variable-Order HMM for Extended Temporal Context: A limitation of the `hmmlearn` implementation is that it only supports first-order HMMs, so predictions consider only the immediately preceding observation. To capture longer-range patterns, we implement a variable-order HMM for the third experiment where inputs are discrete cluster IDs. During training, the model stores probability distributions for contexts up to 50 steps long, as well as all shorter contexts down to first-order. During prediction, it first attempts to match the longest available context and falls back progressively to shorter ones. The variable-order HMM allows the model to generate sequences conditioned on extended musical context, such as phrase-level structure. Furthermore, we adjust the generation process to assign more weights to probabilities

from the local prime sequence. This allows the model to favor patterns unique to each test prime instead of relying solely on the training data.

IV. EVALUATION

This study only focuses on the explicit task: the model generates a continuation for a given prime, and the result is compared against the ground truth continuation. We use two evaluation metrics suggested in the MIREX task: the Cardinality Score and the Pitch Score.

A. Cardinality Score

The Cardinality Score measures how well the predicted continuation matches the rhythmic and structural pattern of the ground truth. The generated continuations are transcribed into sequences of onset and pitch values using `librosa.yin`. We then determine how many note events in the prediction can be aligned with note events in the ground truth under the best possible temporal and pitch shift.

The final score functions similarly to an F1 measure: it increases when the prediction contains the correct note events and decreases when events are missing or incorrect. A higher Cardinality Score indicates that the system successfully reproduced the temporal placement and general melodic contour of the target continuation.

B. Pitch Score

The Pitch Score evaluates how closely the predicted continuation matches the melodic content of the ground truth, regardless of rhythm. Each continuation is converted into a pitch histogram that reflects the distribution of pitch values used. The Pitch Score is then computed as the amount of overlap between the two histograms.

To account for transpositional similarity, the metric is also computed using pitch-class histograms (i.e., pitch modulo 12), so a continuation in the correct key but a different octave is not penalized. A score of 1 indicates complete agreement in pitch content, while 0 indicates no shared melodic material.

V. RESULTS

This paper explored three different modeling strategies for the explicit continuation task defined in Section IV: PCA-reduced spectrograms + Gaussian HMM, Clustered spectrogram + Categorical HMM, Clustered Musical Features + Variable-Order HMM. Across these three iterations, the system evolved from unusable outputs to continuations that were musically interpretable, though still limited in fidelity and structure.

Metric	Attempt 3 (HMM)	MIREX Markov (approx.)
Cardinality Score	0.263 ± 0.208	$\approx 0.27\text{--}0.30$
Pitch Score	0.429 ± 0.196	$\approx 0.45\text{--}0.50$

TABLE I: Attempt 3 vs. MIREX Markov baseline on the PPDD test set (explicit task).

1) *Attempt 1 – PCA-reduced spectrograms + Gaussian HMM:* Modeling PCA-reduced spectrograms (80-D $\rightarrow$ 20-D) directly with a first-order HMM yielded outputs that resembled noise. The absence of pitch or rhythm makes it impossible to compute meaningful Cardinality or Pitch Scores.

2) *Attempt 2 – Clustered spectrogram + Categorical HMM:* Using k-means to quantize spectrogram slices and a first-order HMM to model cluster sequences yielded sporadic melodic hints but overall noisy, low-fidelity outputs. Continuations often got stuck on jittery notes or silence. This is likely because the first-order HMM considers only the immediately preceding observation, preventing it from capturing longer-range melodic or rhythmic context, which in turn makes score computation unreliable.

3) *Attempt 3 (best) – Clustered Musical Features + Variable-Order HMM:* We replaced the first-order HMM with a variable-order model on cluster sequences, which reduced noise and revealed clearer long-term structure in the continuations. By incorporating chroma, RMS energy, and spectral features into the spectrogram slices before clustering, the model generated continuations with improved perceptual quality, including more stable pitches and clearer melodic contours. These improvements in feature engineering and modeling increased both Cardinality and Pitch Scores.

A. Quantitative Evaluation

We focus the quantitative evaluation on Attempt 3, our best-performing configuration. As described in Section ??, we use the MIREX Cardinality Score and Pitch Score for the explicit task on the 15 test primes. The Cardinality Score indicates how many ground-truth note events are correctly reproduced (after optimal translation), while the Pitch Score measures the overlap between the pitch histograms of the predicted and ground truth continuations. Table I reports mean and standard deviation across test pieces, alongside approximate MIREX Markov baseline ranges for reference.

On average, the audio-based HMM performs slightly worse than the symbolic Markov baseline on both metrics, reproducing somewhat fewer correct note events and slightly less accurate pitch content. The relatively large standard deviations reflect substantial variability across primes: some excerpts achieve moderate Cardinality and Pitch Scores, while others remain near zero. This is

consistent with the mixed perceptual quality from our subjective listening.

VI. DISCUSSION

This study demonstrates that an end-to-end audio-to-audio pattern prediction system is feasible using traditional sequence modeling techniques like Hidden Markov Models. By leveraging beat-synchronous feature extraction and clustering, we established a pipeline that operates without symbolic data. However, this audio-to-audio modeling introduces unique challenges that symbolic baselines do not face.

A primary technical limitation observed was the introduction of artifacts during the beat-slicing and reconstruction process. Because the system generates audio by concatenating 16th-note slices based on cluster centroids or exemplars, the resulting waveform often lacks continuity at segment boundaries. Furthermore, the fixed segmentation strategy biased the generation towards short, staccato note events, struggling to reproduce sustained notes or complex phrasing. The reliance on strict tempo normalization also reduces the system's ability to handle expressive timing variations found in natural performance.

Crucially, our results highlight a fundamental mismatch between the goals of audio generation and the existing MIREX evaluation framework. The MIREX Patterns for Prediction task relies on the Cardinality and Pitch Scores, which are inherently symbolic metrics. To evaluate an audio system using these metrics, the generated waveform must first be transcribed back into a symbolic representation. This dependency introduces error into the evaluation process. Standard audio-to-MIDI transcription algorithms are inherently imperfect and assume high-fidelity audio input. In our case, where the generated audio naturally contains slicing artifacts and spectral inconsistencies, the transcription step is liable to misinterpret the signal, producing symbolic sequences that do not accurately represent the musical content of the continuation.

Moreover, these metrics fail to capture the rich acoustic dimensions (e.g., timbre, dynamics, articulation) that motivate the shift to audio modeling in the first place. A generated continuation might be spectrally coherent and texturally interesting but receive a low Pitch Score simply because the transcription step failed to identify a distinct fundamental frequency. Consequently, we argue that the current MIREX evaluation suite is insufficient for audio-to-audio systems. Future research should develop metrics that assess perceptual audio quality and stylistic coherence directly from the audio signal, rather than relying on symbolic transcription.

VII. CONCLUSION

This paper explores a complete audio-to-audio continuation system for the MIREX Patterns for Prediction

task, establishing an HMM baseline for non-symbolic generative modeling. While early attempts using raw spectrograms proved too noisy, the integration of clustered musical features with a Variable-Order HMM yielded structurally coherent musical continuation, despite low audio fidelity.

While the HMM approach offers greater interpretability and reproducibility than advanced deep learning methods, it faces clear limitations in audio fidelity and handling polyphony. Future research could explore deep learning architectures, such as neural audio or diffusion models, to achieve higher-fidelity waveform reconstruction. Studies could also move beyond monophonic piano datasets to polyphonic and multi-instrument contexts to evaluate model performance on more complex inputs. Advancing audio pattern prediction further may require new evaluation standards beyond MIREX, incorporating perceptual listening tests and musicality-centered metrics. These standards could also support interactive systems that enable real-time human-machine musical dialogues.

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